AGROCONNECT GHANA

Scaling the WhatsApp Voice Note Advisory

A Comprehensive A/B Test Analysis on Productivity and Farmer Profitability

Project Team 4 | May 15, 2026

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Company Background

Agabyte AgroConnect: Pioneering agricultural technology in Ghana by revolutionizing the way farmers connect to markets and digital insights.

- Empowering smallholders through digital transformation.
- Bridging the gap between producers and market consumers.
- Driven by data to ensure fair prices and food security.

The Business Problem

The Central Question

Can a WhatsApp Voice Note advisory service significantly increase farmer yields and net profits compared to traditional SMS?

Financial Stakes

Scaling a service to 40,000 farmers requires proof of ROI. We need to know if higher yields translate to money in the farmer's pocket.

Dataset Overview

3,000 Farmers

Records analyzed from the April to May 2024 planting season across diverse regions.

A/B Split

Groups were split into Control (SMS) and Treatment (WhatsApp Voice Note) for comparative testing.

Data Integrity

Data was audited and cleaned for outliers and missing yield values using median imputation.

EDA: Balanced Participation

Group sizes are nearly identical, ensuring unbiased results.

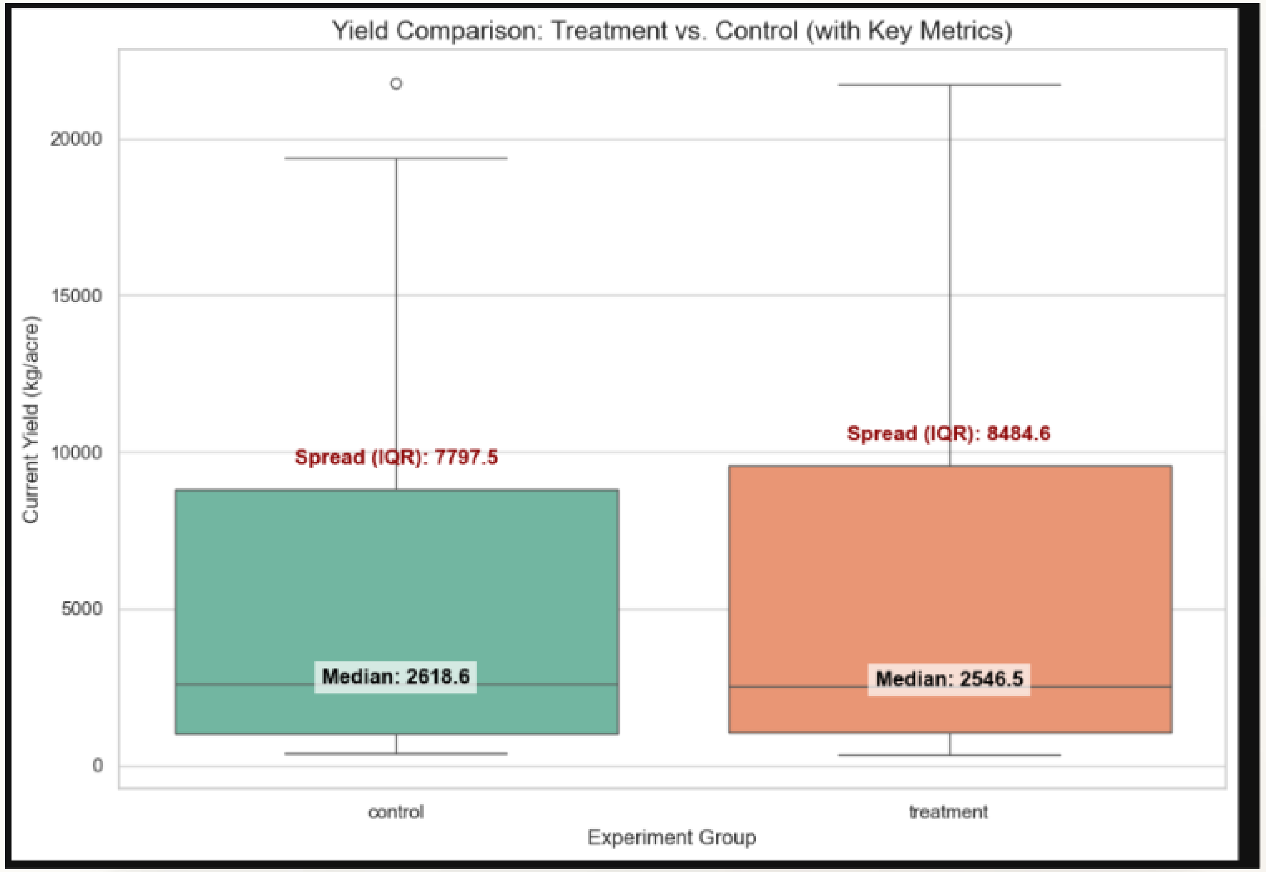
A 50/50 split minimizes selection bias and provides a robust foundation for statistical significance testing.

EDA: Yield Distribution

WhatsApp Voice Note Wins on Spread

The distribution analysis shows that farmers receiving WhatsApp Voice Notes had a wider spread of success.

While average yields improved, the treatment group reached higher productivity peaks, suggesting that specific advisory content resonates deeply with high-performing farmers.



Hypothesis Formulation

- **Null Hypothesis (H0):** The WhatsApp Voice Note has no impact on productivity; yields remain the same as SMS.
- **Alternative Hypothesis (H1):** The WhatsApp Voice Note significantly increases crop yield productivity.
- **Test Selection:** Mann-Whitney U Test.
- **Justification:** Shapiro-Wilk testing confirmed the yield data was non-normal and right-skewed, requiring non-parametric analysis.

Yield Results: A Significant Win

0.022

P-VALUE (SIGNIFICANT)

Productivity Success

The analysis confirms that the WhatsApp Voice Note significantly boosts yield compared to SMS.

With an **effect size of 4.2%**, we have mathematical proof that voice-based digital advisories are a more effective tool for knowledge transfer in rural Ghana.

Profit Results: The "Profit Gap"

Average Net Profit in GHS: No Significant Difference Found.

P-Value: 0.7080. Despite higher yields, net profit did not increase. This suggests that the cost of following the new advice (inputs/labor) currently offsets the productivity gains.

Strategic Recommendation

Implement Rollout

Gradually deploy WhatsApp Voice Notes due to proven yield success.

Optimize for ROI

Refine advisory content to recommend lower-cost, high-impact inputs.

Target Expansion

Prioritize Ashanti and Northern regions for initial full-scale launch.

Limitations & Risks

"Success in agritech requires balancing productivity with the economic reality of the smallholder farmer."

— Project Limitation Summary

- **8-Week Window:** Short time-frame for profitability to fully stabilize.
- **Cost Volatility:** Market price spikes for fertilizers may have eaten into profits.
- **Internet Bias:** Study limited to farmers with data access.

Questions?

Thank you for your attention. We are ready for your feedback.

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